

# **Chapter 8**

## **Addition Reactions of Alkenes**

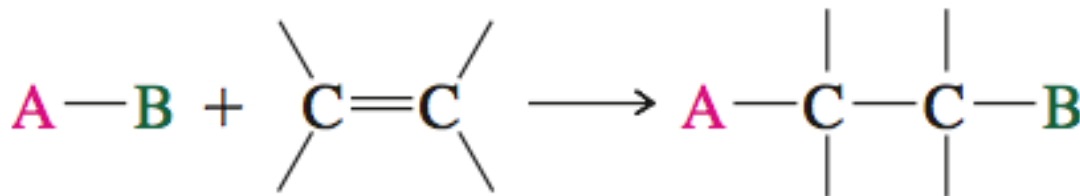
Additional Recommended Problems: Chp8: 25, 28-31, 34, 46-50, 64-66

## CH 8 Learning Objectives

- Predict the product when an alkene react with a hydrogen halides
- Predict the products when alkenes react with HBr/peroxides
- Predict the product when an alkene react with  $\text{H}_2\text{O}/\text{H}^+$
- Predict the product when an alkene undergoes hydroboration/oxidation
- Predict the products when alkenes undergoes oxymercuration/demercuration
- Predict the product when an alkene undergoes hydrogenation
- Predict the product when an alkene reacts with  $\text{Cl}_2$  or  $\text{Br}_2$
- Predict the product when an alkene reacts with  $\text{Cl}_2$  or  $\text{Br}_2$  in the present of water
- Predict the product when an alkene undergoes expoxidiation, with or without water present
- Predict the product when an alkene undergoes ozonolysis
- In all of the above reactions, include effective consideration of reaction orientation (Markovnikov versus anti-Markovnikov orientation), and stereochemistry
- Predict when a reaction will produce achiral versus chiral products
- Predict the correct stereoisomers for stereospecific reactions.
- **Draw detailed logical mechanisms** for alkene reactions with HBr,  $\text{H}_2\text{O}/\text{H}^+$ ,  $\text{Br}_2$ , or  $\text{Br}_2/\text{H}_2\text{O}$ .
- **Use retrosynthetic analysis to solve multi-step synthesis design problems involving alkenes as intermediates or final products**
- Use clues provided by products of reactions such as ozonolysis to determine the structure of an unknown alkene
- Determine the stereochemistry of a starting alkene, given reactants and the product stereochemistry.

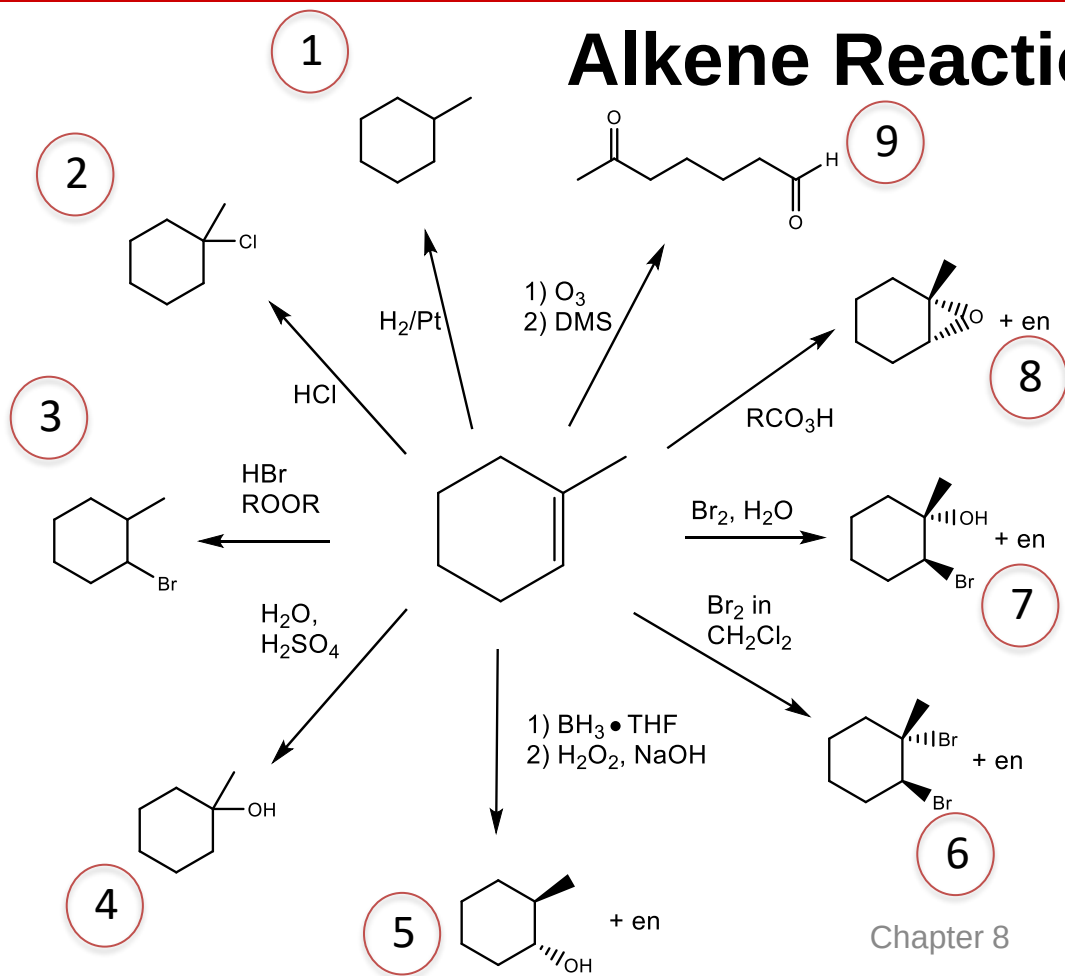
# Additions to Alkenes

- **Additions** are a class of organic reactions involving formation of new bonds to an organic substrate



- A and B in the reagent “add across” the ends of the  $\pi$  system. Often (but not always) the A–B bond is highly polarized
- Note that addition is the opposite of elimination!
- Addition is synthetically valuable because:
  - Readily available alkenes are the reactants
  - Two new  $\sigma$  bonds are formed to the carbons of the alkene

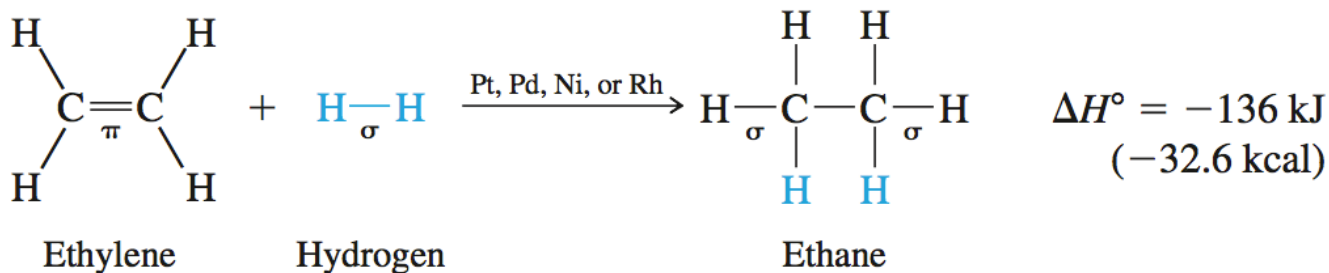
# Alkene Reaction Wheel



1. Hydrogenation
2. Hydrohalogenation (Markovnikov)
3. Hydrohalogenation (anti-Markovnikov)
4. Acid-catalyzed hydration
5. Hydroboration-oxidation
6. Bromination
7. Halohydrin formation
8. Epoxidation
9. Ozonolysis

# 1. Hydrogenation

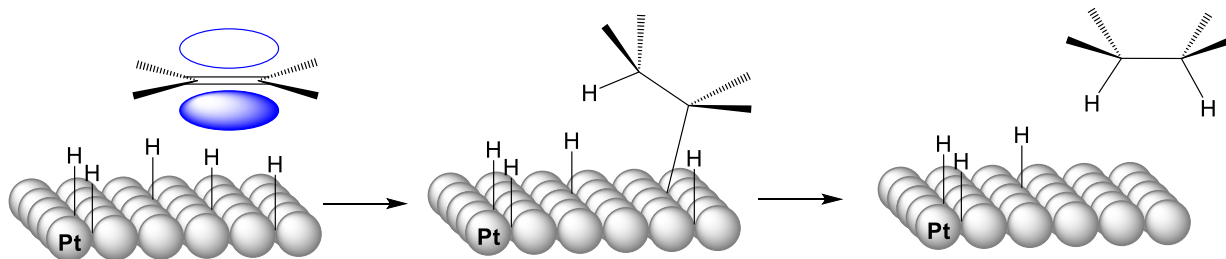
- The addition of H<sub>2</sub> across an alkene is called **hydrogenation**
- A *very* useful method for preparing saturated hydrocarbons from alkenes



- Hydrogenation is universally exothermic
- A metal catalyst is required to activate H<sub>2</sub>; typically the reaction conditions are heterogeneous

# 1. Hydrogenation-1

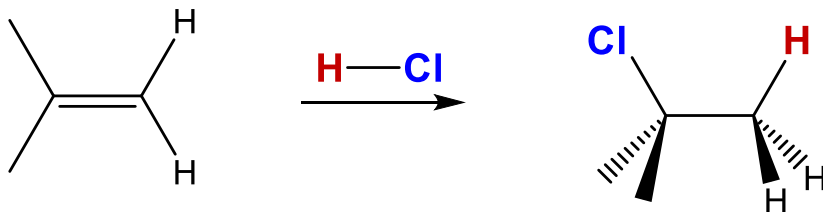
- Mechanism:



- syn*-addition of  $H_2$  across double bond, generates alkane (*reduction of  $C=C$* )
- Heterogenous catalysis,  $H_2$  and alkene adsorbed to surface of metal (Pt common)

## 2. Hydrohalogenation

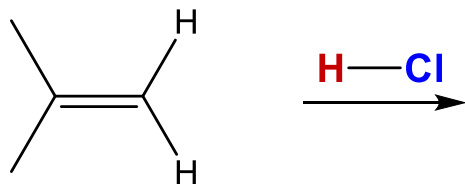
- Reaction:



- Alkenes are nucleophilic and the  $\pi$  bond can act as a Lewis base!
- Treatment with an electrophile (Brønsted or Lewis acid) results in chemical reaction
- The chemical reaction is called electrophilic addition

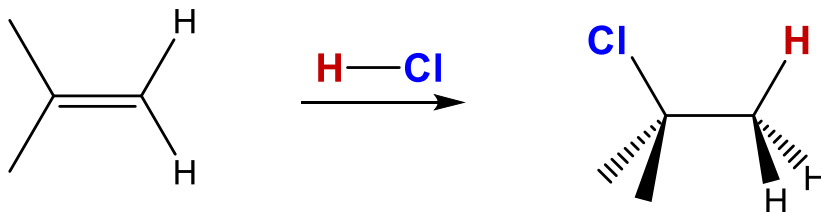
## 2. Hydrohalogenation-1

- Mechanism

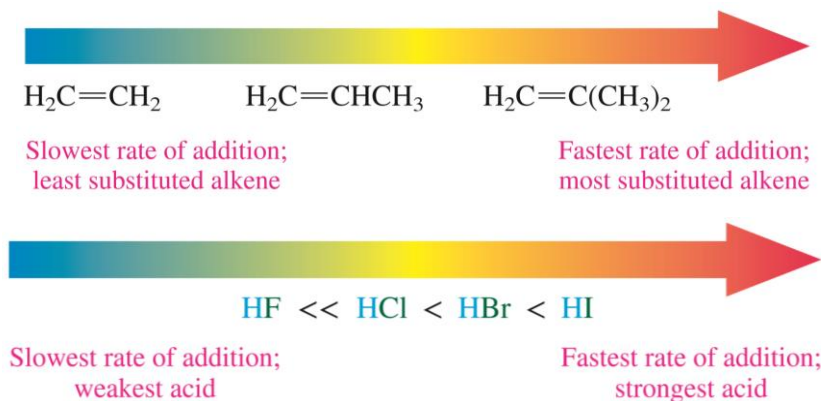


## 2. Hydrohalogenation-2

- Reactivity Trends

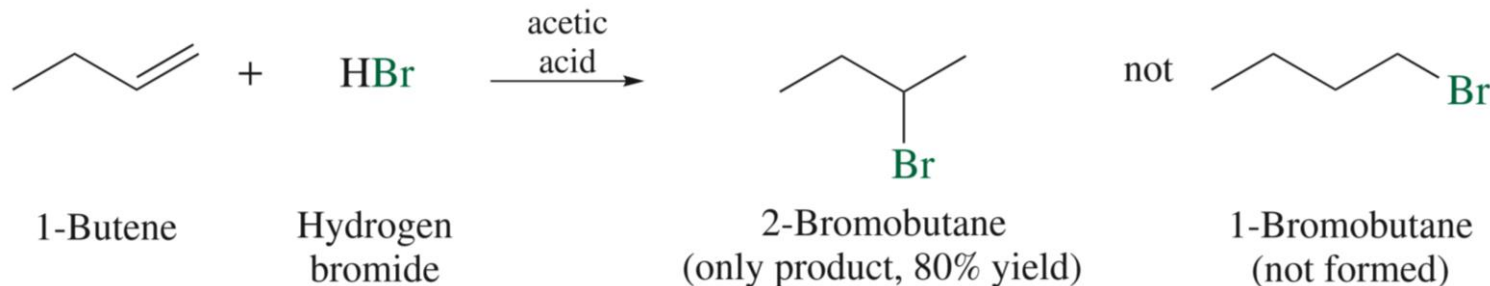


- Reactivity trends:* stronger acids and more substituted alkenes react more rapidly



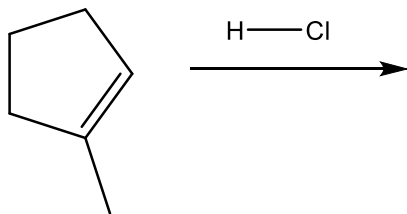
## 2. Hydrohalogenation-3

- Regioselectivity
  - Markovnikov's rule:** hydrogen ends up bound to the carbon with *more hydrogens* in the alkene
    - Most stable carbocation formed, X- adds to most subst. C atom



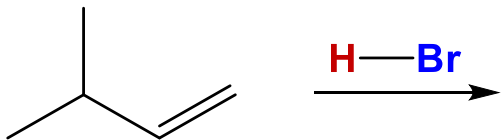
## 2. Hydrohalogenation Example

- Draw the mechanism and show the major product for the following addition reaction.



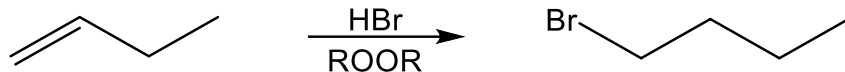
# Rearrangements

- Carbocations often rearrange to form a more stable carbocation (usually 2° to 3°)
- 1,2 shifts are common (hydride *OR* alkyl shift)



### 3. Hydrohalogenation (anti-Markovnikov)

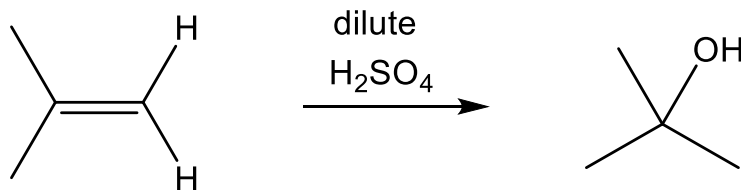
- Reaction:



- Will go over mechanism in CH 10 – radical reaction
- anti*-Markovnikov addition – Br on least substituted carbon.

## 4. Acid Catalyzed Hydration

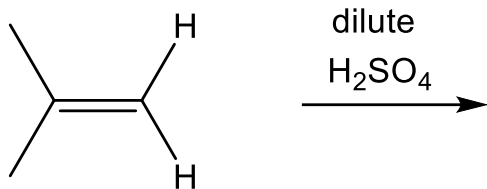
- Reaction:



- In the presence of catalytic acid, H<sub>2</sub>O can add across the carbons of an alkene just like HX
- The product is an alcohol and the mechanism is analogous to electrophilic addition of HX

## 4. Acid Catalyzed Hydration-1

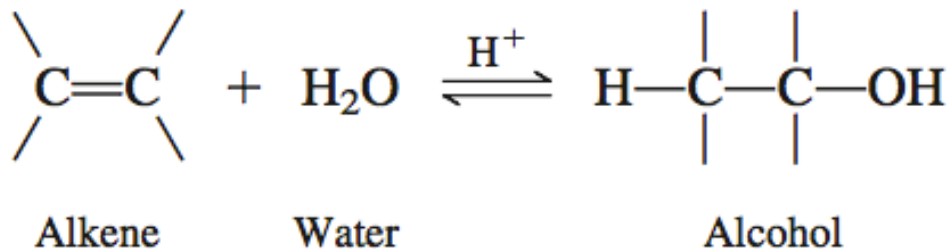
- Mechanism:



- \*Follows Markovnikov's rule

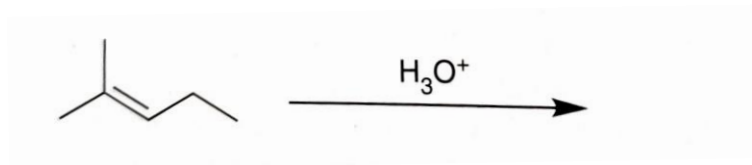
# Hydration is Reversible

- Both *dehydration* (elimination) and *hydration* (addition) can occur, even though they are microscopic reverses of one another!
- Hydration is **reversible**—hydration and dehydration represent a finely balanced equilibrium



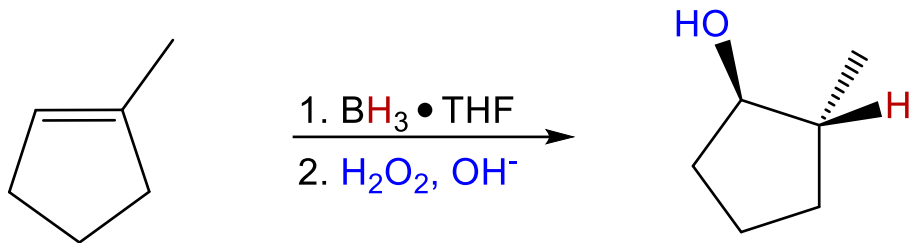
# Hydration Example

- Predict the product for the following reaction and propose a mechanism.



## 5. Hydroboration-Oxidation

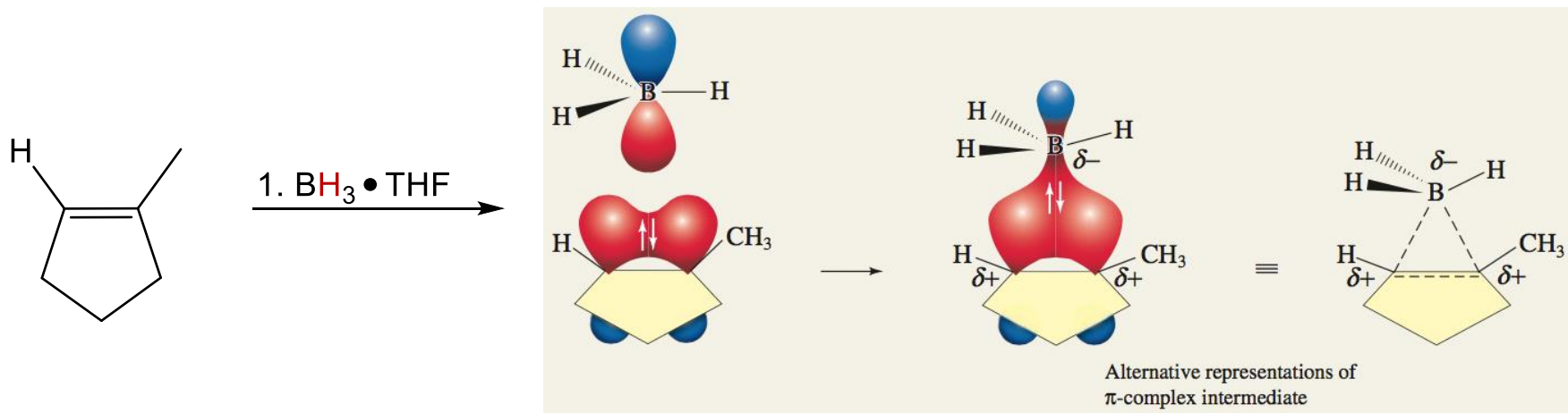
- Reaction:



- Addition of  $\text{-H}$  and  $\text{-OH}$  in an *anti*-Markovnikov fashion.
- Stereospecific and lead to syn addition
- No carbocation formation (concerted; 4-member ring TS)

## 5. Hydroboration-Oxidation-1

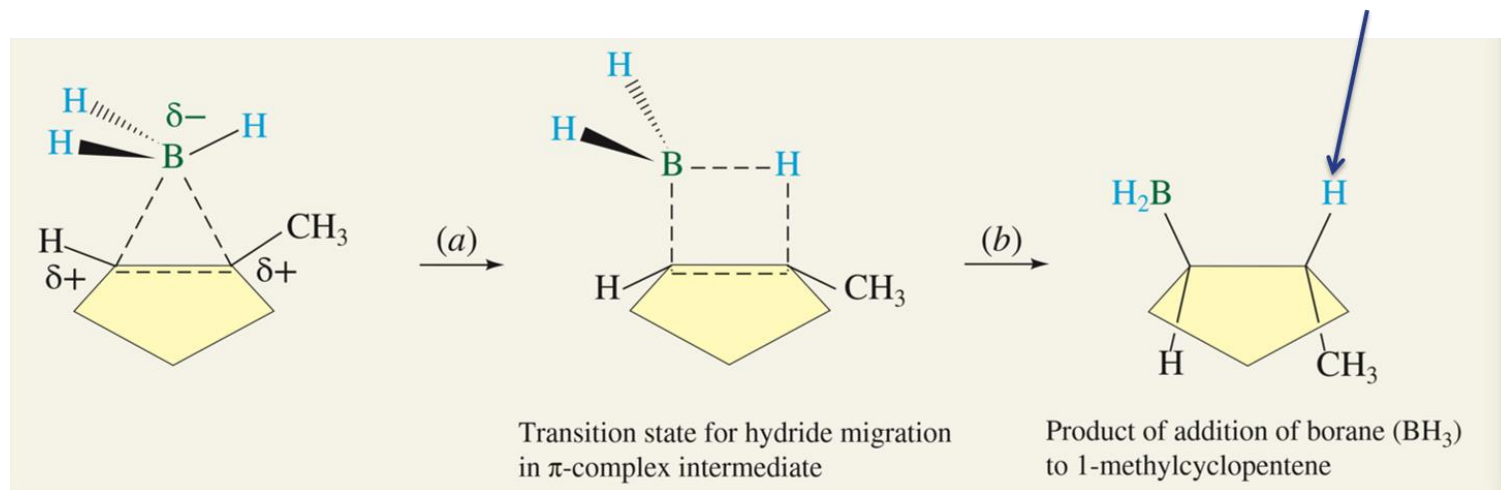
- **Hydroboration Mechanism:**
  - Step 1: the Lewis basic  $\pi$  bond forms a complex with the Lewis acidic boron atom



## 5. Hydroboration-Oxidation-2

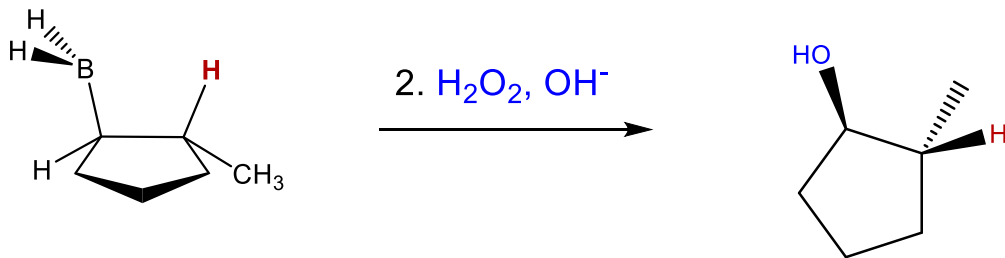
- **Hydroboration Mechanism:**
  - Step 2: H and B add across the bond in a concerted, anti-Markovnikov fashion

Hydrogen forms a bond to the *more substituted* carbon

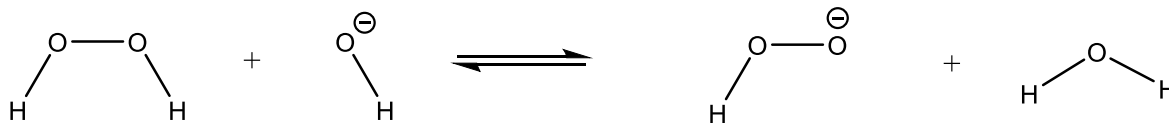


## 5. Hydroboration-Oxidation-3

- Oxidation Mechanism:

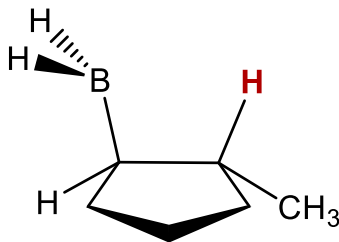


- Step 1:
  - Hydrogen peroxide is converted to its anion in basic solution



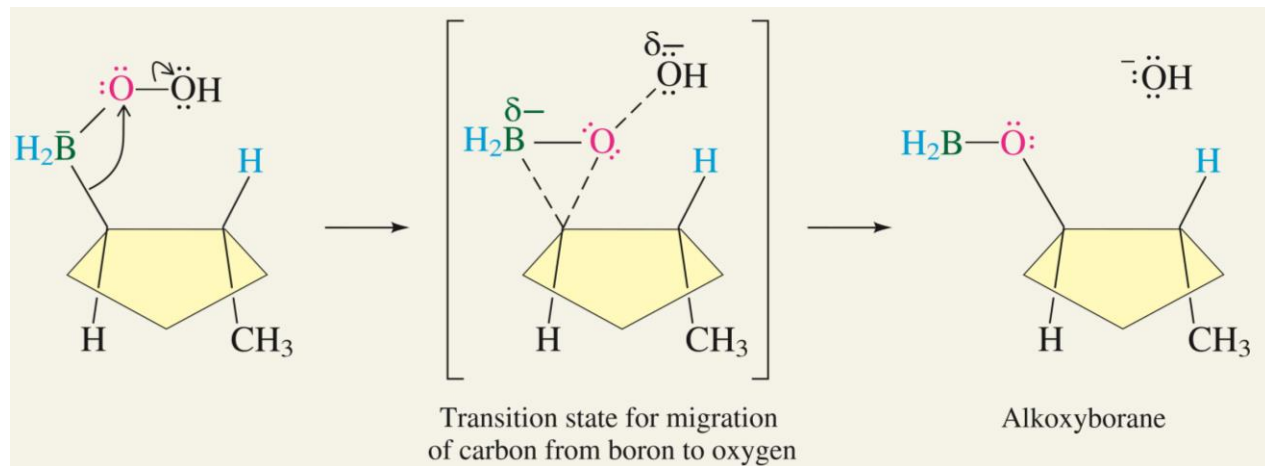
## 5. Hydroboration-Oxidation-3

- **Oxidation Mechanism:**
- Step 2
  - Anion of hydrogen peroxide attacks boron



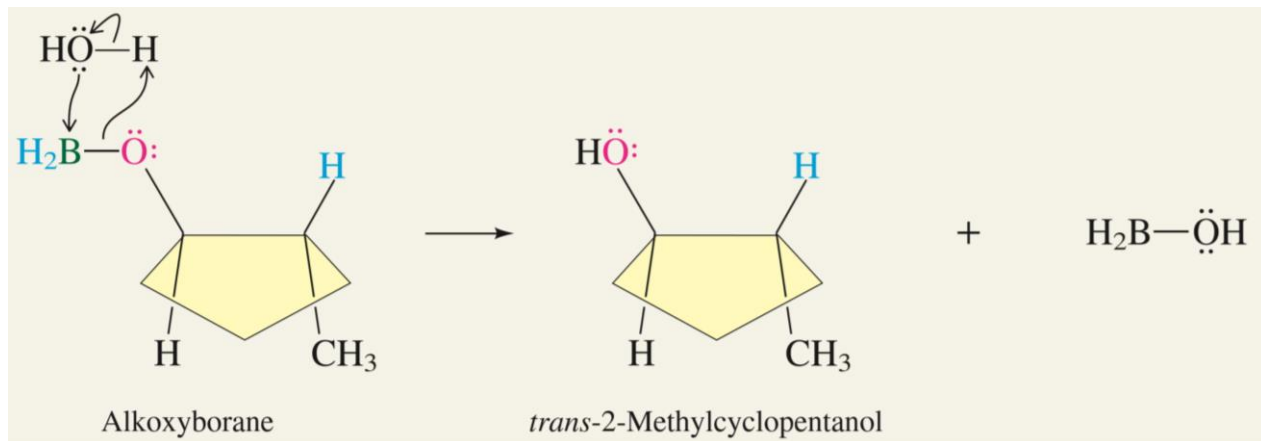
## 5. Hydroboration-Oxidation-4

- **Oxidation Mechanism:**
- Step 3
  - Carbon migrates from boron to oxygen, displacing hydroxide ion. Carbon migrates with the pair of electrons in the C-B bond



## 5. Hydroboration-Oxidation-5

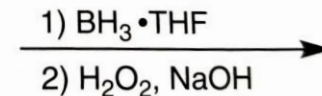
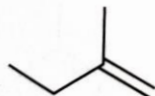
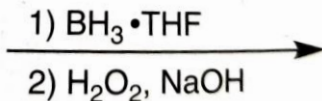
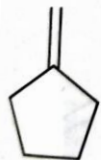
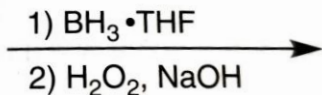
- **Oxidation Mechanism:**
- Step 4
  - Hydrolysis cleaves the B-O bond, yielding the alcohol.



Most importantly, the configuration at the alcohol carbon is retained throughout oxidation.

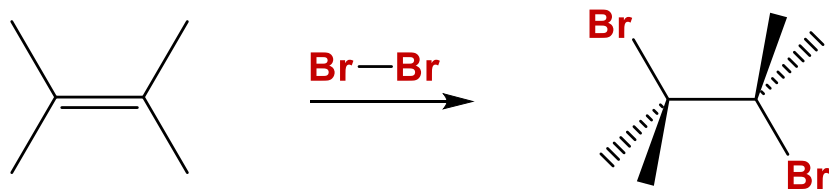
## 5. Hydroboration-Oxidation Examples

- Predict the products for the following reactions.



## 6. Halogenation

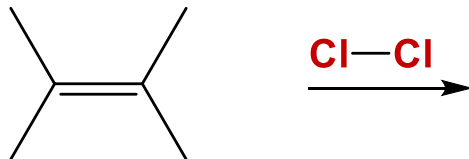
- Reaction:



- anti*-addition of  $\text{X}_2$  ( $\text{Cl}_2$  or  $\text{Br}_2$ ) across double bond
- Mechanism - 3-membered ring with halonium ion, opposite face attack by  $\text{X}^-$

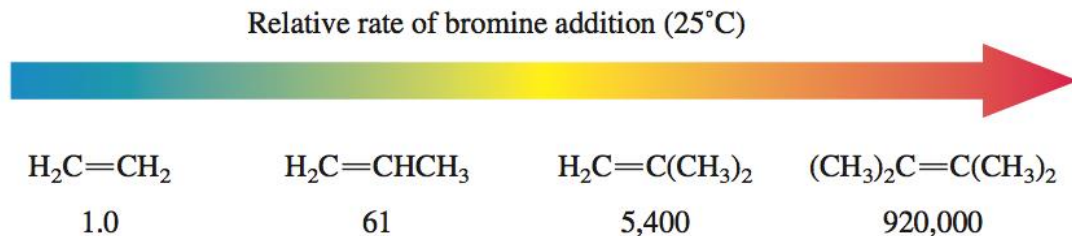
## 6. Halogenation-1

- Mechanism:



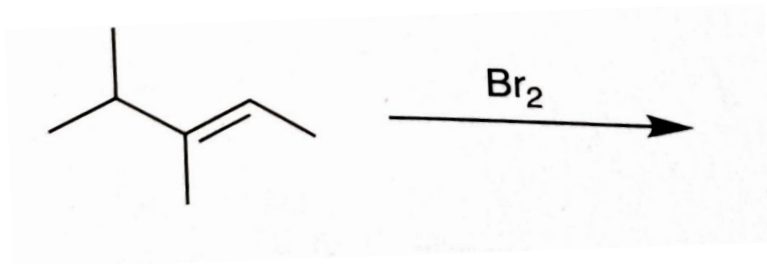
## 6. Halogenation-3

- Halonium Ions
  - Formation of the halonium ion is the RDS; more substituted = faster rate



## 6. Halogenation Examples

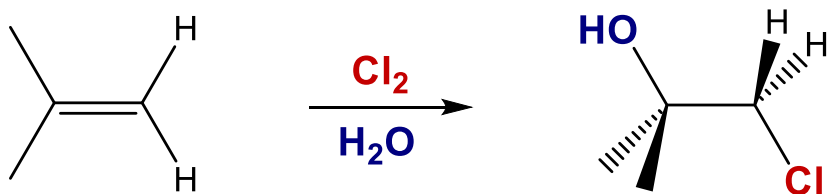
- Draw the mechanism and predict the products for the following reaction.



## 7. Halohydrin formation

### Addition of $X_2$ , $H_2O$

- Reaction:

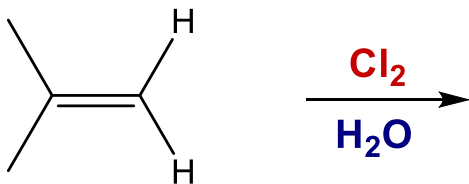


- anti*-addition X and OH, generates halohydrin
- Mechanism – formation of halonium ion, opposite face attack by  $H_2O$  (on most subst. C)

## 7. Halohydrin formation

### Addition of $X_2$ , $H_2O$ -1

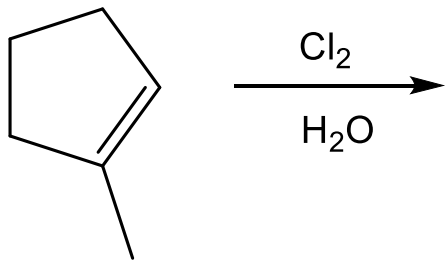
- Mechanism:



## 7. Halohydrin formation

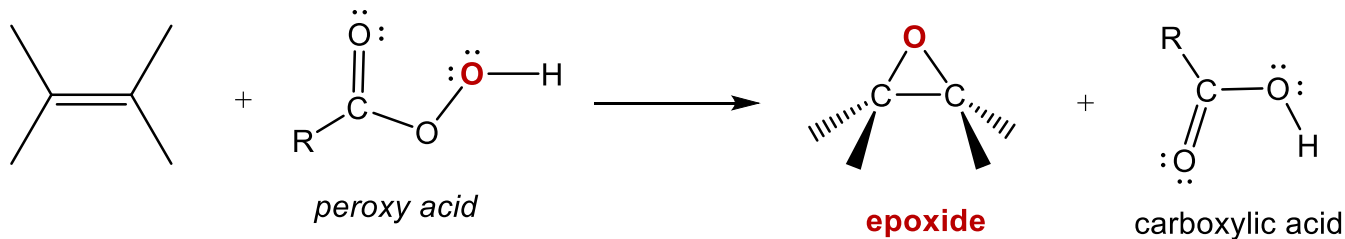
### Addition of $X_2$ , $H_2O$ -2

- Example: Draw the major product(s) for the following reaction.



## 8. Epoxidation

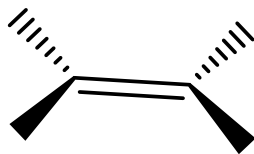
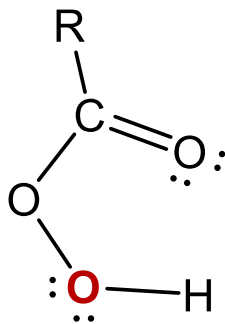
- Reaction:



- syn*-addition using peroxy acids
- Concerted mechanism generates epoxide and carboxylic acid

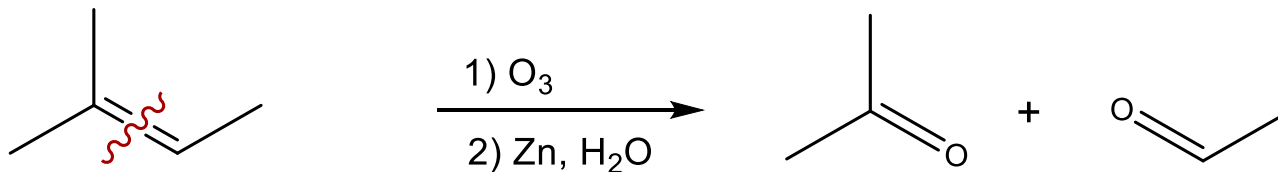
## 8. Epoxidation-1

- Mechanism:



## 9. Ozonolysis ( $O_3$ )

- Reaction:

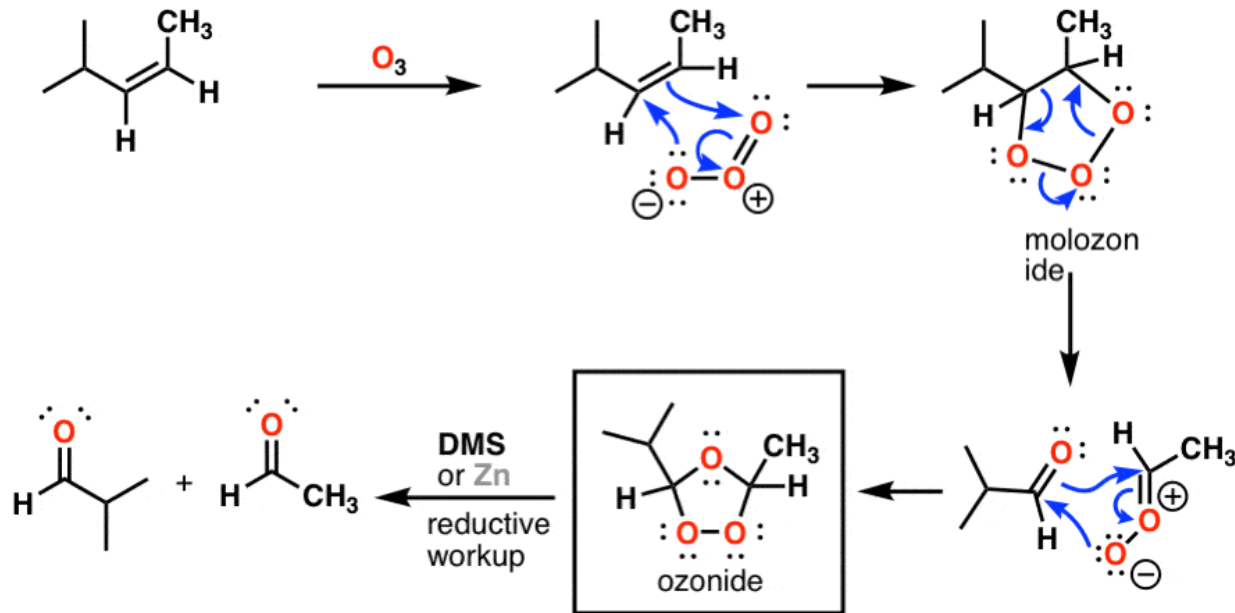


- Cleaves  $C=C$  to form **two**  $C=O$  bonds
- $Zn$  or  $S(CH_3)_2$  utilized in Step 2 to maintain aldehydes
- Ring-opening mechanism

## 9. Ozonolysis ( $\text{O}_3$ )

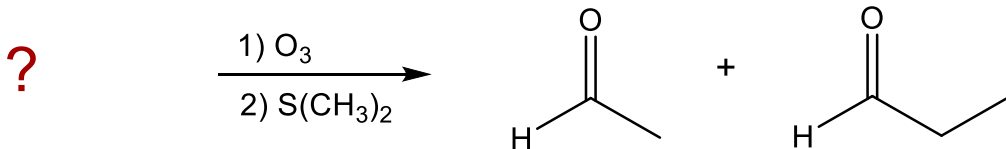
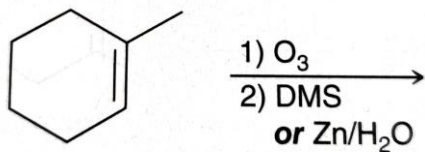
- Mechanism:

How it works: Oxidative cleavage of alkenes

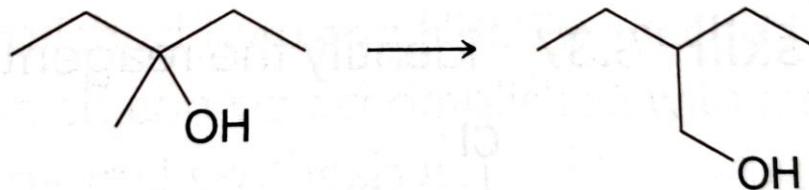
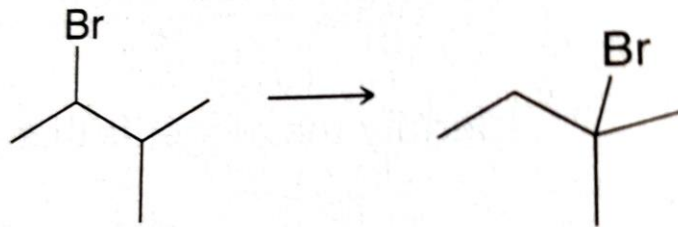


## 9. Ozonolysis ( $\text{O}_3$ ) Example

- Predict the products or reactants for the following reactions.



# How would you accomplish the following transformations?



# Alkene Reaction Wheel

